

AF-FNO Results Tracker

Frozen forward emulators, derivative tests, and scientific decisions

Status date: 23 July 2026 Protocol: AF_FNO_Project_Plan.tex

Current scientific decision. Keep Models A and B frozen as controlled baselines. A0 fails the complete forward gate, and the available 100-day evidence shows that B materially stabilizes A but still loses to persistence for temperature and SSH. The queued one-year A/B evaluations were cancelled before execution: they would characterize already-known failures but would not improve either model. Model C development is now open under the validation-only forward protocol; Model D will be the selected C design plus local-response supervision. The adjoint campaign remains closed until C passes the forward gate.

1 Purpose and reading guide

This document is the scientific ledger. The output tree remains the canonical store for JSON, NPZ, and figures; this tracker selects the decision-relevant results and states their interpretation. Operational job history remains in `Project_tracking.tex`, which is intentionally not duplicated or modified here.

Document	Role
AF_FNO_Project_Plan	Scientific design, Bire-trend benchmark, frozen protocol, and predeclared gates.
Project_tracking	Jobs, configurations, paths, checksums, milestone status, and operational chronology.
Results_tracking	Major figures, numerical findings, failure analysis, and go/no-go decisions.

2 Reference trends from Bire et al.

Bire et al. use a finer and more turbulent configuration, so their numbers are not targets for this 1° tutorial. Their evidence pattern is the reference:

Evidence	Reported trend	AF-FNO test
Map interval	Ten days balanced short predictability and long stability better than five or 30 days.	One frozen ten-day map for A0-D; 10-360-day curves.
Baselines	FNO beat persistence and pointwise climatology before decorrelation; velocity decorrelated first.	Physical-unit RMSE and ACC against both baselines.
Spatial error	Errors concentrated first at the energetic western boundary.	Spatial RMSE, four-cell boundary band, and vertical profiles.
Long dynamics	Exact phase was lost, but integrations stayed bounded and retained characteristic scales.	Stability scalars, spectra, snapshots, and SSH Hovmöller diagrams.
Wind physics	Ten-day-map maximum streamfunction retained an approximately linear wind-amplitude trend.	S1/S0/S2 slope now; inference-only I1/I2 later.
Known failure	A sudden wind change did not reliably steer a state drawn from another regime.	Same-state forcing branches, later checked against switched-wind MITGCM.

3 Common evaluation protocol

All complete packages use the same 45 starts: 15 chronologically fixed starts in each of S0 control, S1 low wind, and S2 high wind. Each frozen ten-day map is rolled for 36 steps. The climatology is a regime-specific pointwise mean computed only from training snapshots; the model has no seasonal forcing, so no unsupported day-of-year baseline is introduced. The packages contain per-member arrays, not just plotted summaries.

- fields: U, V, temperature, SSH, surface speed, SST, PHIHYD at levels 0, 7, and 14, and barotropic streamfunction;
- maps: 10, 60, 180, and 360 days; spectra: 10, 90, 180, and 360 days;
- diagnostics: RMSE, ACC, western-boundary error, vertical drift, spectra, physical regime response, forcing switch, finite/mask checks, and gate scorecard;
- uncertainty display: ensemble mean and 10–90% member range;
- machine-readable evidence: `forward_complete_metrics.json` and `forward_complete_arrays.npz` beside every figure set.

3.1 Pressure postprocessing contract

Pressure is not a learned input or output. Evaluation contract v2 deterministically reconstructs the same MITGCM PHIHYD quantity used by Bire et al.: hydrostatic pressure anomaly divided by ρ_0 , in $\text{m}^2 \text{s}^{-2}$. It applies the configured linear EOS to all 15 predicted temperature levels, uses MITGCM’s finite-difference vertical integration, and adds the linear-free-surface contribution $g\eta$. Bire’s surface, mid-depth, and bottom indices are 0, 7, and 14. Multiplication by $\rho_0 = 999.8 \text{ kg m}^{-3}$ converts the diagnostic to Pa.

Comparison	Level	Max abs. error	RMSE
Surface	0	3.88×10^{-4}	1.48×10^{-5}
Mid-depth	7	1.64×10^{-4}	7.13×10^{-6}
Bottom	14	1.26×10^{-4}	6.63×10^{-6}
All 15 levels	—	3.89×10^{-4}	9.75×10^{-6}

These errors compare the postprocessor with the archived S0 MITGCM PH dump at iteration 2,851,200 over 3,600 wet cells and pass the 10^{-3} tolerance. The report is `outputs/af_fno/pressure_validation_v1/pressure_validation.json`. Pressure adds a Bire-facing physical consistency view, but it is correlated with temperature and SSH and is not presented as three independent prognostic successes.

Existing `forward_complete_v1` outputs are retained as immutable pre-pressure evidence. The saved v1 summaries do not contain every member’s full 15-level forecast at every lead, so complete pressure curves cannot be recovered from them alone. After C freezes, each frozen checkpoint will therefore receive one inference-only replay into a new `forward_complete_v2` directory. This is postprocessing/evaluation, not retraining or model selection.

3.2 Current data and expansion decision

The version-1 chronological split contains 7,530 overlapping ten-day training pairs, 780 validation pairs, and 1,860 inference pairs across S0–S2. Daily starting times are strongly correlated, so these counts are not counts of independent ocean states. There is not yet evidence that raw data volume is the main limitation: B ended at its last scheduled epoch, its spectral term contributed only about 1% of validation loss, and the aggregate 46-channel objective hid the temperature/SSH failures.

Data decision. Do not replace version 1 now. First run Model C learning curves at 25/50/75/100% of training chronology, a stronger small-sample fit, groupwise train/validation errors, autocorrelation estimates, and three-seed spread. Expand only if those diagnostics show a data-limited regime. If approved, create a non-destructive `trajectories_v2` with separated branches from equilibrated checkpoints and target two times the effective independent coverage before considering three times.

4 Previously established ten-day comparison

These values come from the earlier all-inference-pair evaluation and remain useful as a cross-check. Lower than one beats persistence.

Grouped variable	A0 / persistence	A / persistence	B / persistence
U	0.395	0.169	0.179
V	0.912	0.467	0.465
Temperature	4.771	2.052	1.814
SSH	13.612	2.277	2.107

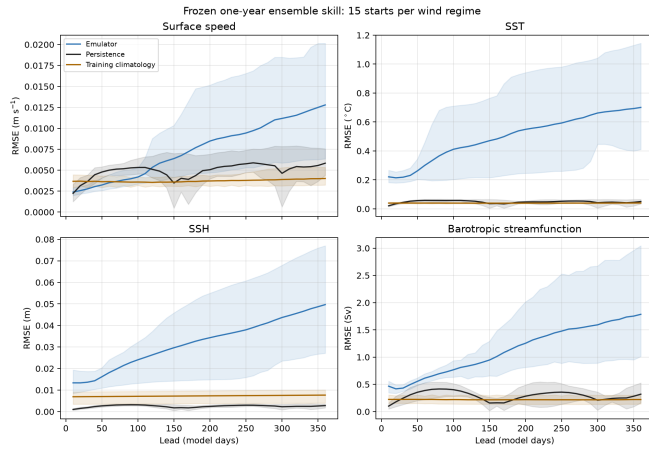
Interpretation. A is unambiguously better than A0 on this held ten-day comparison, especially for velocity. B preserves that short-lead gain and modestly improves temperature and SSH, but both remain

worse than persistence. At 100 days, B's model/persistence ratios are 0.449, 0.997, 8.009, and 9.925 for U, V, temperature, and SSH. Relative to A at the same lead, B reduces those four errors to 0.521, 0.409, 0.450, and 0.405. Thus rollout training works, but the frozen B objective did not solve the slow-field failure.

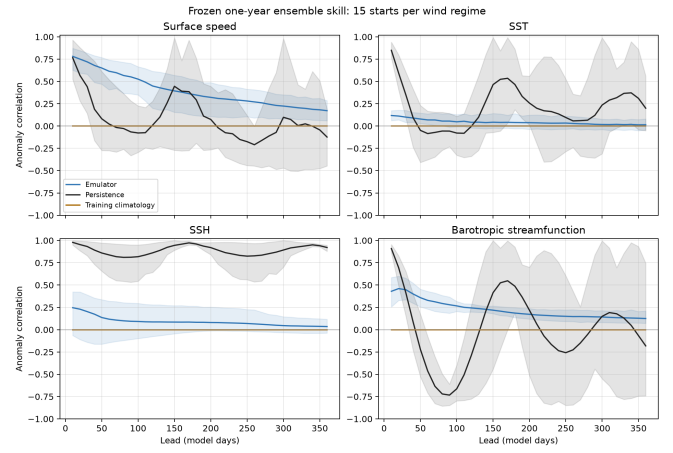
5 A0: adapted Bire architecture

Gate status: Fail. The pre-pressure v1 45-member, 10–360-day package is finished. A0 remained finite, masked, and bounded (maximum normalized wet-state magnitude 18.63) and reproduced the sign and magnitude of the streamfunction–wind slope (model/truth slope 0.897). It failed the ten-day surface-field, 20-day ACC, and one-year spectral gates. Day-360 model/truth spectral energy ratios were 6.48 for surface speed, 96.16 for SST, 281.46 for SSH, and 21.49 for streamfunction.

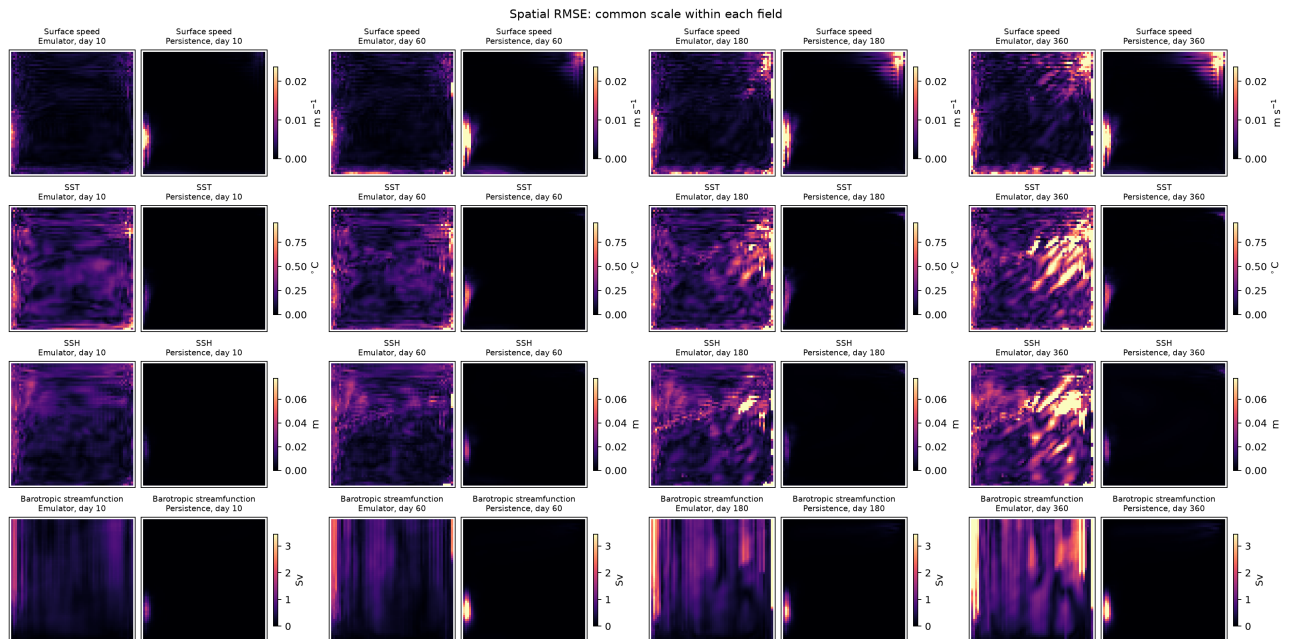
The v1 failure already prevents A0 from passing; pressure-complete v2 will be generated for fair field-by-field comparison, not to reconsider or retrain A0.



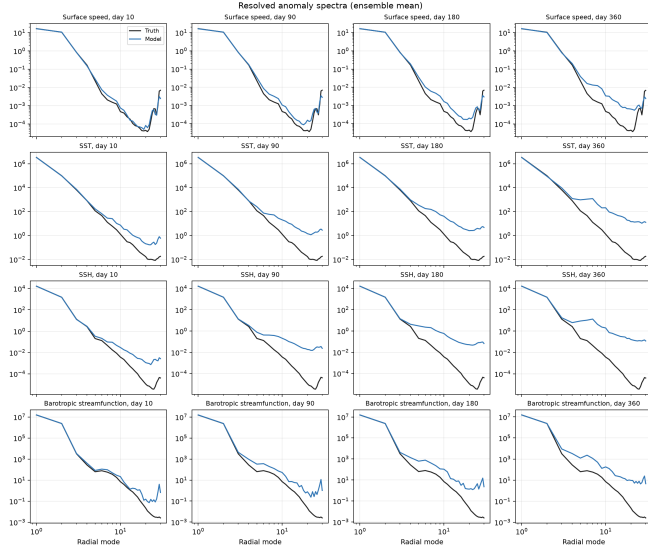
Physical RMSE against persistence and training climatology.



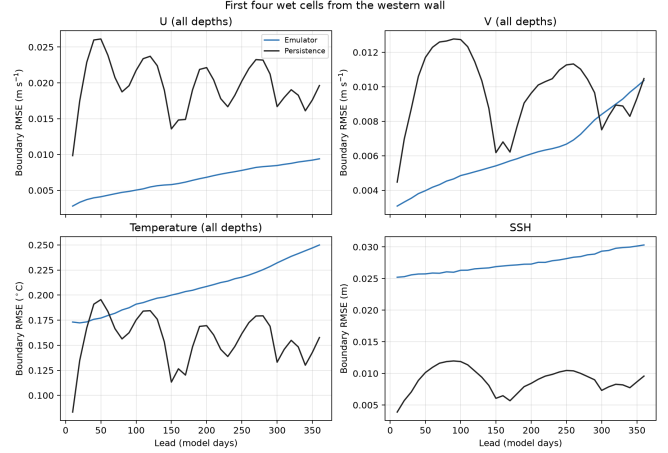
Pointwise-climatology anomaly correlation.



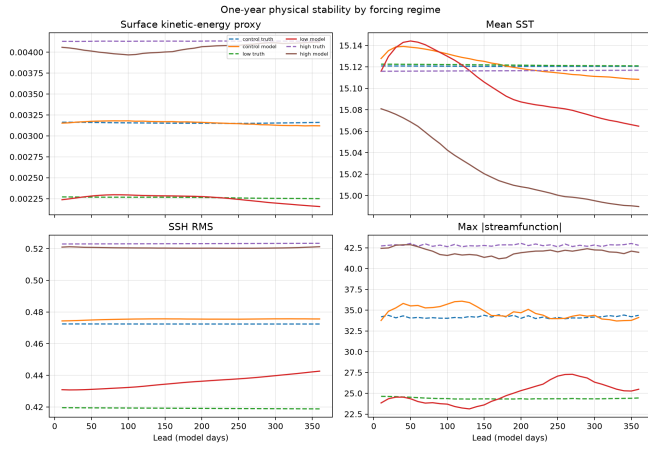
A0 spatial RMSE for surface speed, SST, SSH, and streamfunction. Emulator and persistence share a scale within each field.



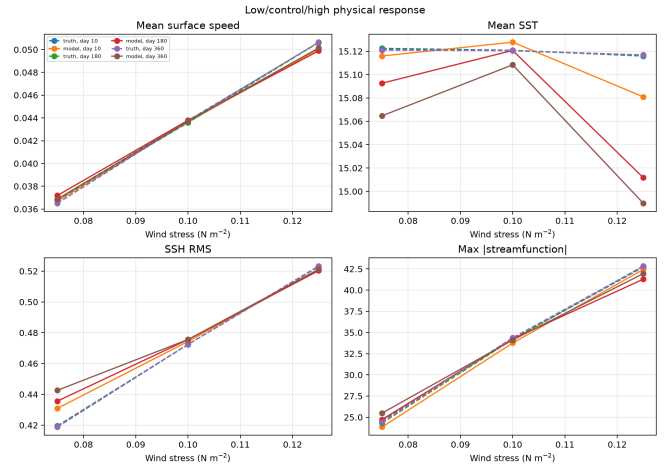
Ensemble-mean resolved spectra.



First four wet cells from the western wall.



One-year physical stability by wind regime.



Low/control/high response at fixed lead times.

6 Model A: modern state-residual baseline

Gate status: Fail on available mandatory evidence; complete one-year characterization not run. A is frozen and reproducible, but its ten-day temperature and SSH errors are 2.052 and 2.277 times persistence, so it cannot pass the forward gate. Its 100-day ratios rise to about 17.8 and 24.5. The queued complete evaluation was cancelled with zero runtime because it could describe, but not reverse, this disqualifying result. Missing panels below are retained as explicit planned-output placeholders rather than fabricated results.

Planned output

../outputs/af_fno/A/forward_complete_v2/forward_rmse_vs_lead.png

Physical RMSE against persistence and training climatology.

Planned output

../outputs/af_fno/A/forward_complete_v2/forward_acc_vs_lead.png

Pointwise-climatology anomaly correlation.

Planned output; not generated

../outputs/af_fno/A/forward_complete_v2/forward_spatial_rmse_maps.png

Model A spatial RMSE for surface speed, SST, SSH, and streamfunction.

Planned output

../outputs/af_fno/A/forward_complete_v2/forward_spectra.png

Ensemble-mean resolved spectra.

Planned output

../outputs/af_fno/A/forward_complete_v2/forward_vertical_drift.png

Day-360 mean and variance drift by depth.

Planned output

../outputs/af_fno/A/forward_complete_v2/forward_stability_by_regime.png

One-year physical stability by wind regime.

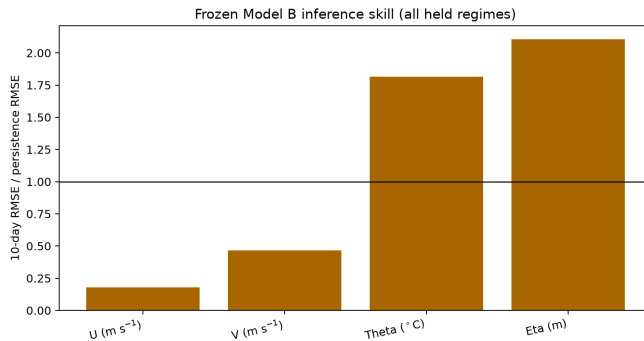
Planned output

../outputs/af_fno/A/forward_complete_v2/forward_forcing_switch_response.png

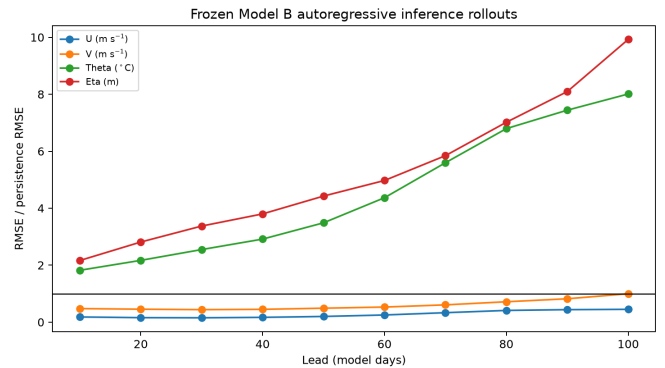
Provisional same-state forcing-switch response.

7 Model B: frozen rollout/spectral/boundary ablation

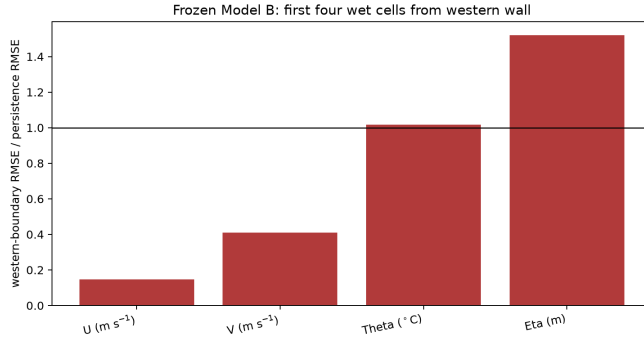
Gate status: Fail on available mandatory evidence; retain as a controlled ablation. B greatly reduces A's 100-day error but still has ten-day temperature and SSH errors 1.814 and 2.107 times persistence. Its frozen validation objective was also poorly balanced: state, rollout, spectral, and boundary terms contributed approximately 41.6%, 39.0%, 1.0%, and 18.3% of the total. This is evidence to redesign C's objective, not permission to retune B.



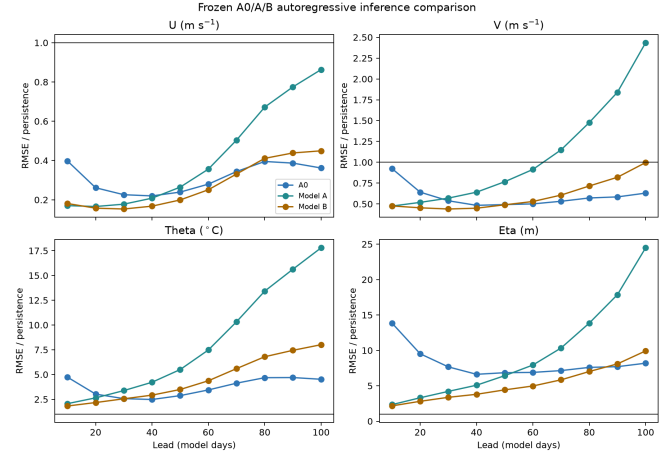
Ten-day Model B skill by grouped variable.



Ten-to-100-day rollout against persistence.



Western-boundary skill reveals where the local and boundary losses help or fail.



Controlled A0/A/B rollout comparison.

8 Models C and D: revised forward/response separation

Model	Frozen role	Decision rule
C	Best forward emulator under the bounded search in the project plan	Select only on validation. Every U/V/temperature/SSH ten-day ratio must beat persistence; then rank worst-group error and 30/90/180-day physics scores. Run the complete one-year inference package only after freezing.
D	Selected C design plus local perturbation-response loss	Preserve C's ten-day state error within 5%, lower blind response error by at least 25%, and improve projected-gradient angle without adjoint supervision.

8.1 Model C stage C0: architecture and data audit

Status: C0/C1a complete; C1b rejected; C1c running. Corrected training-only job 284850 completed in 42 seconds; version-2 job 284857 added the data-justified (24, 16) candidate and completed in 33 seconds. Both verified dataset metadata SHA-256 `cbcf2938f458...ca10847` and recorded that validation, inference, I1/I2, response, and adjoint data were not read. Complex-safe CPU calibration job 285116 completed in 4m39s; its report froze the loss scale without opening validation. Pending GPU replica 284858 was cancelled. Job 285192 passed all memorization/reload criteria except temperature/SSH increment skill; job 285265 changes only the duration from 160 to 320 epochs and remains training-only.

Criticism of B / starting C Implemented C0/C correction

Fifteen U, 15 V, 15 temperature, and one SSH channel were aggregated in one relative norm.	Equal mean of U, V, temperature, and SSH group losses; every group is reported separately.
B’s spectrum compares hard-masked full states, for which persistence is already strong and the land discontinuity leaks energy.	Compare ten-day increments, crop the exact 60×60 wet rectangle, remove the spatial mean, and apply a Hann taper before radial binning.
Slow-field increments are small relative to normalized state magnitude.	Scale each channel’s increment error with its RMS computed from training pairs only.
The 16-mode choice had not been tested with NeuralOperator’s exact real-FFT convention.	Audit (12, 12), (16, 16), (16, 24), (24, 16), (24, 24), interpreting the last declared count as $m_x/2 + 1$ stored real-FFT columns.
Width 64 could improve capacity but multiplies cost.	Start at 630,262 parameters (width 32); test 2,443,542 parameters (width 64) only if fit/capacity evidence warrants it.

Modes	U increment	V increment	Temperature increment	SSH increment
(16, 16)	8.8%	24.0%	67.8%	85.6%
(16, 24)	9.0%	25.6%	68.1%	86.8%
(24, 16)	12.5%	32.9%	70.2%	89.0%
(24, 24)	13.5%	38.3%	71.7%	90.9%

C1a term	Raw loss	Gradient norm	Frozen weight	Weighted grad./state
State	0.03018	1.965	1	1.000
Increment	1.89450	830.5	0.001	0.423
Rollout	0.06463	6.484	0.15	0.495
Spectrum	38.4904	40,763	10^{-5}	0.207
Boundary	0.07825	7.411	0.065	0.245

Interpretation. All (16, 16) full-state fractions exceed 99.6%, so the slow-field failure is not evidence that full states need a larger global truncation. Velocity changes are predominantly higher frequency and must rely on the local branch; 24 meridional modes help more than 24 zonal modes. The normalized ten-day increment RMS varies by a factor of 147 across channels, confirming the need for training-only per-channel scaling. Increment-RMS autocorrelation leaves about 290 effective U, 392 V, 278 temperature, and 398 SSH samples across the three regimes; low-wind S1 is the most correlated. Version 1 is adequate for the first bounded search, but raw pair counts substantially overstate independence.

The C1a automatic proposal clipped increment and spectral coefficients at 0.01. It was rejected because those values would still yield weighted gradient ratios about 4.23 and 207 relative to state. Rounded *unclipped* coefficients instead meet the declared targets closely, as the table shows. Thus 10^{-5} spectral weight is not tantamount to removing spectra: the raw spectral gradient is about 20,700 times the state

gradient. No selected C or D result exists yet; C1b must show that this balanced objective can memorize every physical group and reload exactly. Intermediate-wind, response-inference, and adjoint archives remain sealed. If C’s learning curves establish a data limitation, the expanded dataset will be versioned as v2; existing v1 data and conclusions will not be replaced.

8.2 Model C stage C1b: rejected slow-field increment gate

LR	U/pers.	V/pers.	T/pers.	SSH/pers.	State max	Rollout
0.0010	0.226	0.404	1.802	2.888	0.0271	0.0314
0.0005	0.254	0.442	1.285	2.032	0.0304	0.0312

Decision. Job 285192 completed in 58m30s and is scientifically rejected, not a runtime failure. Both attempts reduced total loss by about 97.7%, spectral loss by about 99.1%, and passed state, rollout, and boundary thresholds. Only temperature and SSH ten-day increments remained worse than persistence. Persistence NRMSE on this exact subset is U/V/temperature/SSH 0.968/0.954/0.981/0.948; therefore the failure is not an artifact of assuming an exact baseline of one.

The 0.0005 attempt reached its best total at epoch 160. Stage C1c therefore changes only duration to 320 epochs before any loss or width change. The revised runner records the exact subset persistence baseline, full learning curves, and a bitwise-reload-tested diagnostic checkpoint even on rejection. Job 285265 remains training-only; validation and inference are sealed.

9 Forward gate and decision

Model	Status	Current evidence	Scientific role
A0	Fail	Complete gate: short-skill, ACC, and spectra fail; boundedness and wind slope pass.	Adapted transfer reference
A	Fail	Temperature/SSH lose to persistence at 10 days; unstable error growth through 100 days.	Modern one-step control
B	Fail	Rollout loss materially helps A, but temperature/SSH still lose at 10 and 100 days.	Frozen loss ablation
C	Pending	C0/C1a are complete; C1b rejected slow-field increment skill and controlled-duration C1c job 285265 is pending. No candidate has been validated.	Must open forward gate
D	Pending	Blocked until C freezes and response data pass linearity checks.	Response/adjoint model
Complete adjoint campaign may open			No

10 Adjoint results layout (sealed until the forward gate)

When the forward gate passes, this tracker will add the following paired A/B/C/D evidence, with C versus D as the primary response-supervision comparison:

Result family	Decision-relevant displays and data
Perturbation validity	Central-difference linearity, plus/minus symmetry, Taylor slopes, and noise floor by direction family and region.
JVP response	True/model/difference maps, response scatter, relative error, cosine, gain, boundary distance, depth, and horizon.
Adjoint fields	Signed maps, depth slices, vertical profiles, full/projected angle and error, magnitude, sign overlap, and spectra/coherence.
Consistency	Emulator duality $q^T J v = v^T J^T q$, AD versus finite differences, and ten-to-30-day composition.
Control utility	Full objective-versus-step curves, predicted and actual ΔJ , chosen step, and constraint residual.

11 Artifact index

Artifact	Canonical location
A0 earlier evaluation	outputs/af_fno/A0/final_v1/ and outputs/af_fno/A0/paper_style_v1/
A0 pre-pressure complete package	outputs/af_fno/A0/forward_complete_v1/
A0 pressure-complete package	Planned outputs/af_fno/A0/forward_complete_v2/
Pressure validation	outputs/af_fno/pressure_validation_v1/pressure_validation.json
Model A earlier evaluation	outputs/af_fno/A/final_v1/
Model A pressure-complete package	outputs/af_fno/A/forward_complete_v2/; deliberately not generated yet
Model B existing package	outputs/af_fno/B/final_v1/
Model B pressure-complete package	outputs/af_fno/B/forward_complete_v2/; deliberately not generated yet
Model C C0/C1a evidence	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/training_diagnostics_v2/ and /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/loss_calibration_cpu_v1/
Model C frozen loss manifest	config/model_c_loss_v1.json
Model C C1b gate	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_v1/
Model C C1c duration gate	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_duration_v2/
Models C and D outputs	Planned versioned roots outputs/af_fno/C/ and outputs/af_fno/D/
Frozen criteria	docs/AF_FNO_Project_Plan.tex

12 Decision log

Date	Decision	Evidence and consequence
2026-07-21	Freeze A0	Adapted architecture learned a ten-day signal but failed persistence for temperature and SSH; retain as transfer baseline.
2026-07-21	Freeze Model A	Better than A0 for all four grouped errors, but earlier 100-day temperature/SSH rollouts were poor; build B.
2026-07-22	Expand forward gate	Bire paper audit showed missing climatology, one-year, spectra, boundary, propagation, and forcing-response evidence; regenerate A0/A before adjoints.
2026-07-22	Separate ledgers	Keep operations in <code>Project_tracking</code> ; create this results-and-decisions document with output-tree links.
2026-07-23	Keep B; insert C	Duruisseaux-guided audit found that 16 modes are plausible, but loss balance, group aggregation, and incomplete optimization are stronger current failure explanations. B stays frozen; C receives the bounded forward search; response supervision moves to D.
2026-07-23	Cancel queued evaluations	Jobs 284816 (A) and 284817 (B) were pending with zero runtime and were cancelled. Existing checkpoints and 100-day evidence are preserved; final uniform packages are deferred until C is frozen.
2026-07-23	Retain data v1	Raw pair count is ample for diagnosis but effective independence is unknown. Require learning curves and autocorrelation/seed evidence before a non-destructive v2 expansion.
2026-07-23	Complete Model C C0	Jobs 284850/284857 established a 147-fold increment-scale range, low effective sample counts, and meridionally anisotropic velocity-increment energy. Added (24, 16), retained the local branch, and started training-only gradient calibration job 284858.
2026-07-23	Add CPU calibration replica	GPU job 284858 remained pending for resources. Because C1a is not GPU-specific science, submitted eight-core job 284860. It failed safely before output because float32 overflowed while accumulating a global gradient norm. Float64 fixed overflow, but live job 284864 showed that casting complex gradients to real discarded their imaginary component; it was cancelled before output. A regression test now covers large real and complex gradients using float64/complex128 only for diagnostic reduction.
2026-07-23	Freeze Model C loss scale	Complex-safe job 285116 completed in 4m39s. Rejected the clipped automatic proposal: 0.01 increment/spectral weights would still dominate. Froze rounded unclipped coefficients 0.001/0.15/10 ⁻⁵ /0.065, giving weighted gradient ratios 0.423/0.495/0.207/0.245 relative to state. Cancelled pending GPU replica 284858 with zero runtime and recorded report hash 3406c9104dd6...592e469.
2026-07-23	Start Model C C1b	Added groupwise memorization thresholds and exact three-step reload verification. Fixed initialization and shuffled batch order across the bounded learning-rate fallback so only learning rate changes. Eleven Model C tests and all checks passed; CPU job 285192 is running on the same sealed 96 training rollouts.

Date	Decision	Evidence and consequence
2026-07-23	Reject Model C C1b	Job 285192 passed total, spectral, state, rollout, and boundary gates but failed temperature/SSH increment skill. At learning rate 0.0005 their exact persistence ratios were 1.285/2.032; U/V passed at 0.254/0.442. This is a specific slow-field failure, not an aggregate or runtime failure.
2026-07-23	Start Model C C1c	Because the better attempt reached its best total at the final epoch, test undertraining before changing weights or width. Job 285265 changes only 160 to 320 epochs at learning rate 0.0005. Revised diagnostics preserve subset persistence, complete learning curves, and a reload-tested best checkpoint even if rejected.
2026-07-23	Add pressure evaluation	Implemented the exact linear-EOS, finite-difference MITGCM PHIHYD postprocessor and validated it against the archived S0 PH dump: all-level wet-cell maximum error 3.89×10^{-4} and RMSE $9.75 \times 10^{-6} \text{ m}^2 \text{ s}^{-2}$. Pressure remains derived rather than learned. Contract v2 adds Bire's levels 0/7/14 to RMSE, ACC, maps, snapshots, spectra, boundary diagnostics, and gates. Preserve v1; replay frozen inference once after C freezes because v1 did not save full ensemble states at every lead.